Jay Prajapati

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Professional Summary

Mechanical Engineer in Training (EIT – APEGA) with a Master of Mechanical & Manufacturing Engineering (University of Calgary) and hands-on oil and gas engineering experience, specialising in FEA-based mechanical analysis, mechanical integrity assessment, piping stress engineering, and technical documentation across concurrent industrial client projects. At Zachry, the approach has always been to dig into the mechanics of why equipment behaves the way it does under load — not just report findings, but understand the physics driving them and translate that into clear, actionable engineering recommendations. Brings ANSYS, SolidWorks, Python, MATLAB, ASME and API code knowledge, ABSA AB-520 regulatory experience, a Lean Six Sigma Green Belt, and a track record of producing engineering deliverables that operations and integrity teams can act on.

Experience

Zachry Integrity Engineering Ltd. – Engineering Specialist May 2025 – Present

Zachry Integrity Engineering provides specialized mechanical integrity and engineering assessment services for refinery and petrochemical assets across North America. In my role, I support engineering evaluations of pressure equipment and piping systems, helping clients make technically sound decisions related to equipment integrity, repair planning, and continued operation.

- Supported API 579-1 Level 3 Fitness-for-Service (FFS) assessments for refinery assets, including vacuum towers, coke drums, fractionators, nozzles, cyclic piping systems, and non-standard piping configurations.
- Performed static, dynamic, and fatigue finite element analyses using ANSYS Workbench and APDL to evaluate structural response, localized stresses, deformation, and fatigue-sensitive regions.
- Developed finite element models to evaluate non-standard jacketed pipe cross configurations and support Stress Intensification Factor (SIF) assessments using ASME B31.3 and ASME B31J principles where standard values were unavailable.
- Reviewed client drawings, equipment records, inspection findings, and thermography data to prepare engineering calculations, technical reports, and mechanical integrity assessments.
- Collaborated with multidisciplinary engineering teams and client personnel to deliver technically sound recommendations supporting equipment reliability, repair planning, and continued operation.

Prime Toolings – Propulsion System Design Intern Sep 2024 – Jan 2025

Prime Toolings specializes in the design and manufacturing of precision tooling and engineering solutions for aerospace and industrial applications. During my internship, I supported propulsion-system design activities involving CAD modelling, engineering analysis, and simulation-driven design improvements.

- Designed and refined propulsion-system components and layouts using CAD tools, considering aerodynamic performance, structural requirements, and manufacturability.
- Conducted CFD simulations to evaluate airflow behaviour and compare design iterations, supporting engineering decisions throughout the development process.
- Supported turbine blade and propulsion component development by evaluating geometry, material selection, and structural performance.
- Prepared engineering models, technical documentation, and simulation results to support design reviews and project discussions.
- Collaborated with engineers throughout the design process to deliver assigned project work within established timelines.

Circle K – Assistant Manager Feb 2023 – Present

Circle K is one of North America's largest convenience retail chains, serving millions of customers through a network of company-owned and franchised stores. As an Assistant Manager, I supported daily store operations while leading a team in a fast-paced customer-focused environment.

- Supervised day-to-day store operations, ensuring high standards of customer service, inventory management, cash handling, and operational efficiency.
- Led, trained, and supported a team of employees by coordinating work schedules, delegating responsibilities, and maintaining a collaborative work environment.
- Developed an Excel VBA-based inventory tracking system that improved inventory visibility and contributed to a reduction in stock shortages.
- Helped improve operational performance by supporting initiatives that contributed to increased sales, enhanced workplace safety, and more efficient inventory management.
- Strengthened leadership, communication, problem-solving, and decision-making skills while balancing operational priorities in a high-volume retail environment.

Brahmastra Aerospace – Aerospace Vehicle Design Project Trainee Nov 2022 – Jan 2023

Brahmastra Aerospace provided project-based exposure to aerospace vehicle design, CAD modelling, and computational fluid dynamics. During this training experience, I worked on aircraft and propulsion-related design concepts using CATIA V5 and CFD tools.

- Designed an aircraft wing with internal support structures using CATIA V5, considering aerodynamic geometry and structural arrangement.
- Performed CFD analyses at multiple angles of attack to study airflow behaviour and evaluate aerodynamic performance.
- Developed a simplified rocket model and conducted simulations to examine stability, airflow, and thrust-related characteristics.
- Modelled a jet-engine turbine blade using CAD techniques, considering aerodynamic form and component geometry.
- Documented design assumptions, simulation observations, and project findings as part of the technical training activities.

Certifications

Certificate in Advance Bluebeam - Certificate

- Completed advanced training on Bluebeam Revu, covering architectural markup tools, quantity takeoffs, estimation, document comparison, and PDF-based engineering coordination workflows. (Student ID: AC20788338).

Process Improvement Foundations - Certificate

- Completed professional training covering business process improvement, analyzing operational workflows, identifying process bottlenecks, and applying quality optimization frameworks.

Revit for Mechanical and Plumbing Design Professional - Certificate

- Completed professional Revit MEP training covering HVAC, plumbing, and mechanical system modelling, BIM coordination, and engineering drawing production for commercial building projects.

Lean Six Sigma Green Belt - Certificate

- Completed structured continuous improvement, KPI development, performance monitoring, and data-driven problem solving methodology training.

Reading HVAC, Plumbing and other Drawings & Schematics - Certificate

- Completed formal P&ID, HVAC, plumbing, and engineering drawing interpretation training.

Mechanical Engineering Mastery Series: HVAC Engineering 101 - Certificate

- Completed HVAC engineering fundamentals including thermal load calculations, equipment selection, duct and hydronic system design, and energy efficiency considerations.

Advanced CREO Parametric Training for Product Design and Development - Certificate

- Completed advanced parametric modelling, assembly design, GD&T documentation, and manufacturing drawing production training.

AutoCAD: 3D Modeling for Mechanical Designs - Certificate

- Completed professional course covering advanced 3D mechanical part modeling, parametric sketching, assembly design operations, and 3D drawing setups in AutoCAD.

Reading HVAC Drawings and Blueprints - Certificate

- Successfully completed interactive course training on HVAC trade prints, schematic parsing, and blueprint interpretation standards.

Industrial Innovation through IIOT - Certificate

- Completed specialized training covering Industrial Internet of Things (IIoT), exploring industrial innovation, smart connected system architectures, and smart manufacturing integration.

Introduction to Commercial Systems - Certificate

- Completed interactive course training covering commercial systems installation, operation, and maintenance guidelines under accredited trade standards.

The Science of Effective Communication: Decoding Human Interaction - Certificate

- Completed professional development course covering critical communication frameworks, presentation delivery, decoding interpersonal dynamics, and soft skills leadership training.

Leadership & Projects

Structural Stress & Boundary Condition Analysis of a Gantry Crane - Project

- Performed a structural FEA of a gantry crane with W18×60 beams and HSS columns in SolidWorks. Applied pin-roller boundary conditions to allow natural expansion, calculating a maximum bending stress of 86.81 MPa (Factor of Safety = 2.88 against ASTM A36) and mid-span deflection of 6.84 mm compliant with standard L/600 serviceability limits.

Fitness-for-Service Assessment of Refinery Pressure Equipment (API 579-1 / ASME FFS-1 Level 3)

- Led mechanical integrity assessment for a Vacuum Tower and Coker Main Fractionator, performing API 579-1 FFS evaluations under ASME BPVC VIII and delivering repair and operating specifications accepted by client operations and integrity teams.

SIFs Using FEA for Non-Standard Piping Components (ASME B31J / ABSA AB-520)

- Developed ASME B31.3-compliant Stress Intensification Factors for non-standard jacketed pipe cross configurations using custom ANSYS FEA models, producing defensible design specifications registered under ABSA AB-520 requirements.

Bottom Flange Design of a Coke Drum and Mechanical Analysis

- Performed mechanical design and analysis for a coke drum bottom flange under cyclic operating and transient loads, conducting thickness sensitivity studies using thermography data to verify pressure integrity and support capital repair planning.

Differential Pressure Assessment of Catalyst Beds and Outlet Screens

- Assessed process equipment performance across catalyst beds and outlet screens using vendor drawings and operating data, modelling tray geometry and identifying loading effects to support equipment engineering decisions.

Anchor Bolt Vibration & Resonance Analysis

- Assessed dynamic loading behaviour, resonance conditions, and long-term fatigue risk of anchor bolt connections on mechanical equipment foundations, producing reliability evaluations that informed maintenance planning and corrective action decisions.

Alberta Pipeline Integrity & Proximity GIS Risk Analysis (QGIS)

- Developed an automated geospatial proximity model in QGIS mapping Alberta's pipeline networks relative to watercourses, road crossings, and high-consequence areas (HCAs) under NAD83 datum to prioritize pipeline direct assessment programs.

Random Vibrational Analysis of a Shock Absorber - Project

- Developed a 3D solid-element FEA model of a shock absorber assembly in ANSYS, applying PSD acceleration loading across 5–50 Hz with local mesh refinement to evaluate directional deformation and equivalent von Mises stress under random vibration conditions.

3D CAD Model of a 4-Cylinder Engine Assembly -- SolidWorks - Project

- Designed a fully constrained 4-cylinder engine assembly in SolidWorks, producing fabrication-ready 2D drawings with GD&T callouts through precision parametric modelling, assembly constraints, and manufacturability-focused design.

3D CAD Model of Engine Blower -- CREO - Project

- Developed a 3D engine blower model in CREO Parametric, focusing on precision parametric modelling and manufacturability, resulting in a 15% improvement in blower efficiency.

3D CAD Model of Bench Vise Assembly -- AutoCAD - Project

- Designed a detailed 3D bench vise assembly in AutoCAD with GD&T-annotated technical drawings, focusing on component precision, assembly constraints, and manufacturability.

Aerodynamic Analysis of Finite Span 3D Wing Performance using ANSYS, Python and MATLAB - Project

- Performed 3D CFD simulation and aerodynamic performance analysis using ANSYS, Python, and MATLAB, producing validated results through systematic mesh convergence studies and cross-method comparison.

CFD Analysis of NACA 4412 Airfoil with Grid Independence and Adaptive Meshing - Project

- Conducted systematic CFD simulation in ANSYS Fluent, achieving grid-converged aerodynamic performance data through mesh convergence studies and turbulence model validation.

Jet Noise Reduction Analysis using a Chevron Nozzle - Project

- Conducted aeroacoustic simulations analyzing jet noise reduction using a chevron sawtooth nozzle in ANSYS Fluent and SolidWorks, resolving acoustic power level (dB) dissipation and turbulent kinetic energy fields.

Heat Transfer & Flow Dynamics in a Rectangular Channel - Project

- Investigated heat transfer mechanisms and water fluid dynamics inside a rectangular channel design using ANSYS CFD simulation models under constant 310 K thermal wall boundary conditions.

Energy-Efficient HVAC System Design for Retail Space Using Revit - Project

- Designed a complete HVAC system for a commercial retail building in Revit MEP, performing heating and cooling load calculations, equipment selection, and coordinated system layout under ASHRAE-referenced design methodology.

HAZOP Analysis of Sulphur Recovery Unit for Process Safety and Risk Mitigation - Project

- Conducted systematic process hazard identification and risk assessment for a Sulphur Recovery Unit, documenting hazard scenarios, integrity threats, and corrective actions in a formal HAZOP report.

Education

Schulich School of Engineering, University of Calgary – Master of Mechanical & Manufacturing Engineering

Gujarat Technological University – Bachelor of Aeronautical Engineering

Courses: Finite Element Analysis, Computational Fluid Dynamics, Thermodynamics, Engineering Design, Advanced Material Engineering, Project Management, Aerodynamics, Manufacturing Optimization

Concentration: Mechanical Design, Engineering Analysis, Mechanical Integrity, Pressure Equipment Design, Stress Analysis

Skills

Technical Skills: FEA, Mechanical Integrity Evaluation, API 579-1 FFS, ASME BPVC VIII (Div 1 & 2), ASME B31.3, ASME B31J, ASME B16.5, ASME Sec II Part D, ASME PCC-1 Appendix O, ABSA AB-520, Piping Stress Analysis, Root-Cause Analysis, Failure Analysis, Vibration & Dynamic Analysis, Random Vibration (PSD), CFD, HVAC System Design, ASHRAE 90.1, GD&T, Material Selection, Engineering Calculations, P&ID Reading, HAZOP Participation, Technical Documentation

Software Knowledge: ANSYS Workbench & APDL, ANSYS Fluent, SolidWorks, AutoCAD, Revit MEP, PTC CREO Parametric, CATIA V5, ABAQUS, Bluebeam Revu, QGIS, MATLAB, Python, Microsoft Excel (Advanced, VBA), Power Query

Soft Skills: Analytical Problem-Solving, Technical Communication, Multi-discipline Coordination, Attention to Detail, Continuous Improvement, Independent Work, Safety-First Mindset, Valid Class 5 Driver's Licence